

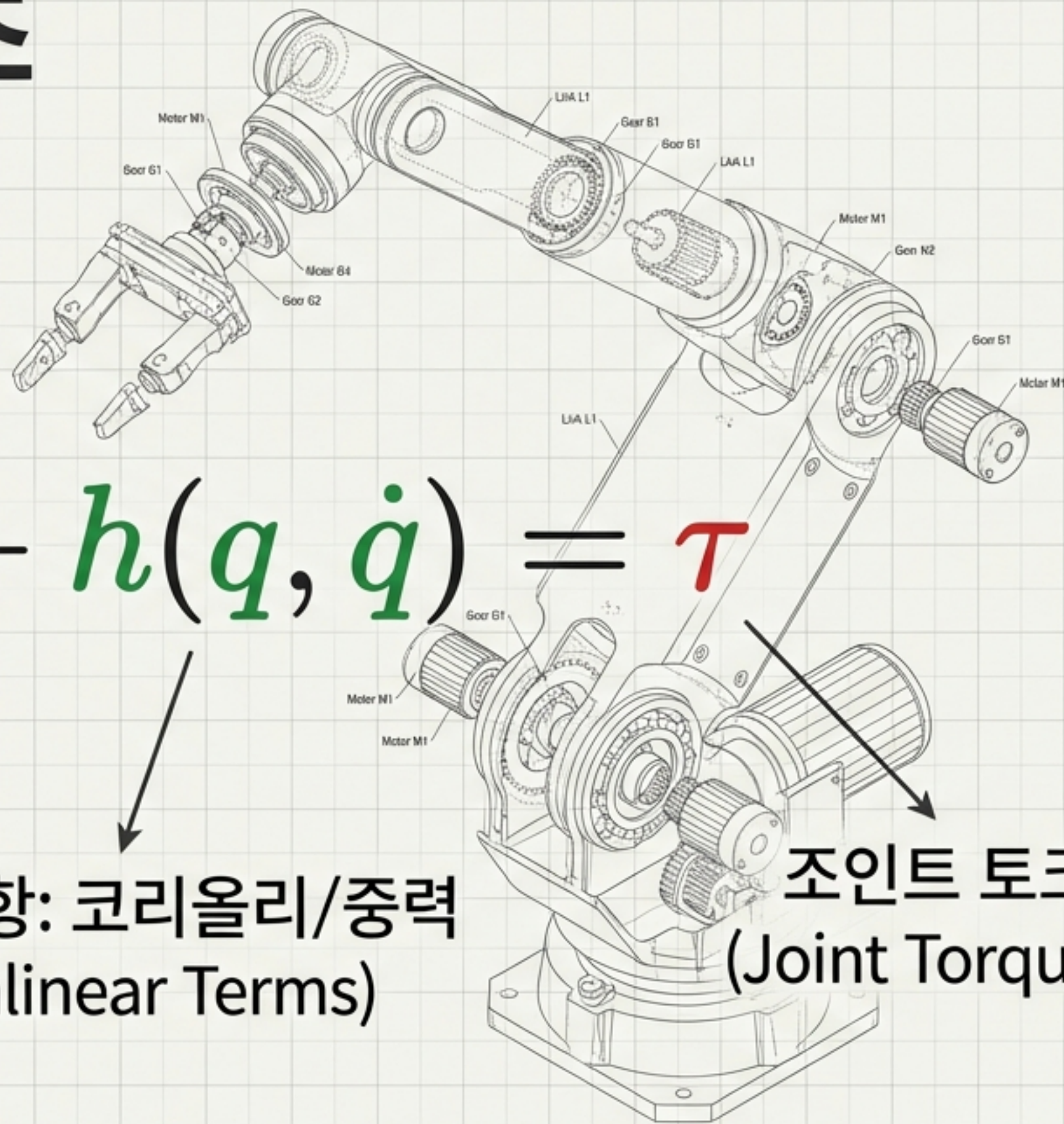
로봇 동역학의 기초

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau$$

질량 행렬
(Mass Matrix)

비선형 항: 코리올리/중력
(Nonlinear Terms)

조인트 토크
(Joint Torques)



제어의 가정과 현실

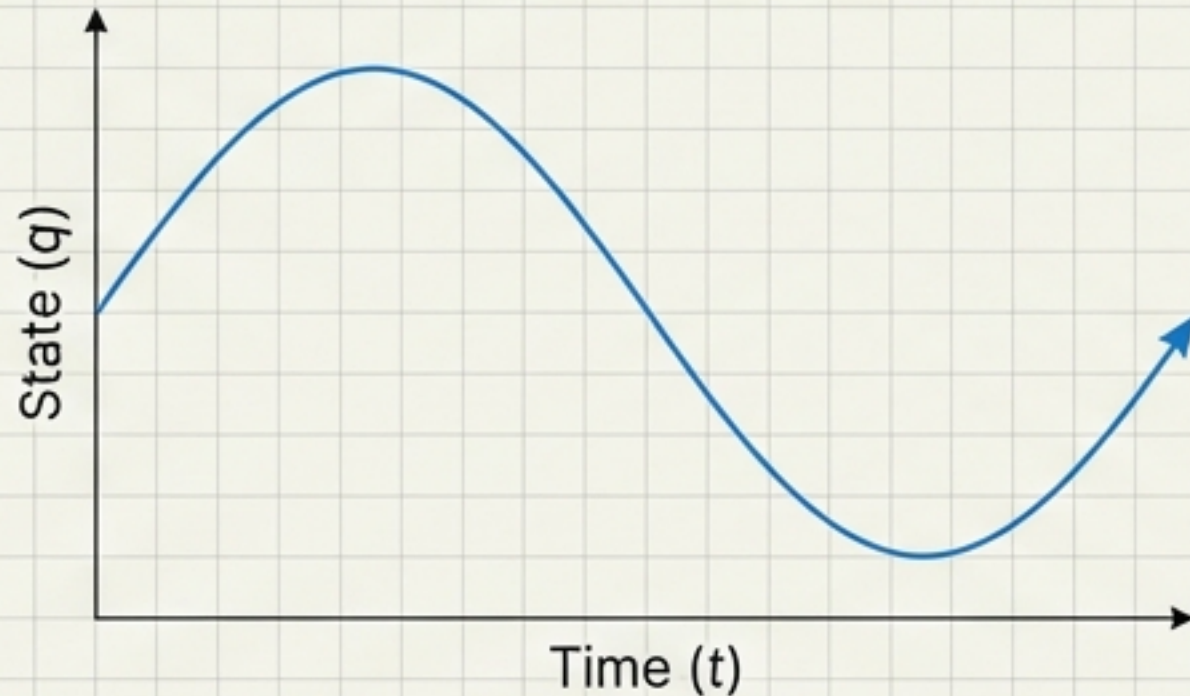
이상적 가정 (Ideal Assumptions)

- 상태 측정 가능 (Measurable State):

$$q, \dot{q}$$

- 완벽한 모델 (Perfect Model):

$$M(q), h(q, \dot{q})$$



현실의 문제 (Reality)

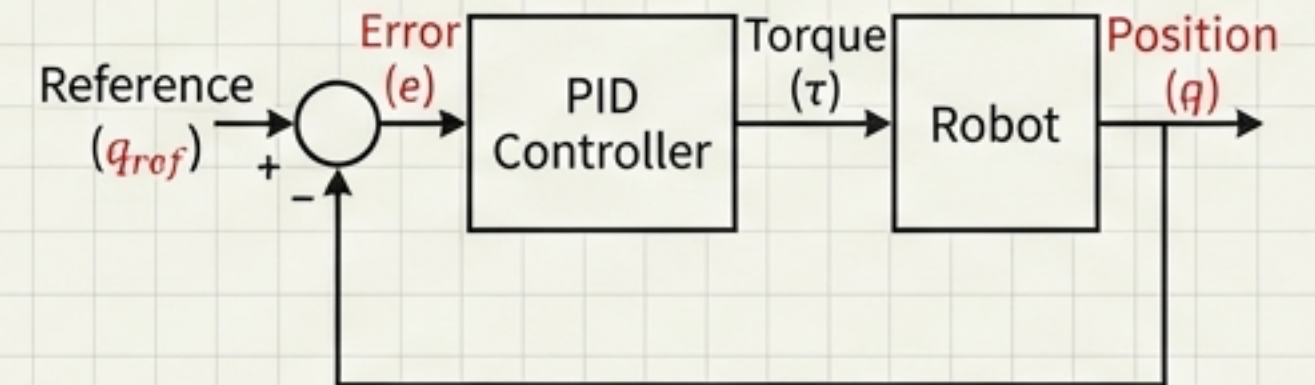
- 마찰 (Friction): 쿨롱 마찰 & 비선형성
- 모델 불확실성 (Model Uncertainty): 파라미터 오차



목표: 참조 궤적 $q_{ref}(t)$ 추종

조인트 공간 제어: PID 제어

$$\tau = K_P e + K_D \dot{e} + K_I \int e dt$$



$$e = q_{ref} - q \quad \dot{e} = \dot{q}_{ref} - \dot{q} \quad K_P, K_D, K_I : \text{대각 게인 행렬 (Diagonal Gain Matrices)}$$

PID 제어의 정상 상태 해석

가정 (Assumption): $\dot{\mathbf{q}} = 0$, $\ddot{\mathbf{q}} = 0$ (Steady State)

폐루프 동역학 (Closed Loop Dynamics): $g(\mathbf{q}) = K_P \mathbf{e} + K_I \int \mathbf{e} dt$

$$\frac{d}{dt} (g(\mathbf{q})) = 0 \quad \frac{d}{dt} (K_P \mathbf{e}) = 0$$

$$\frac{d}{dt} \left(K_I \int \mathbf{e} dt \right) = K_I \mathbf{e}$$

$$\text{Result: } 0 = K_I \mathbf{e} \Rightarrow \mathbf{e} = 0$$

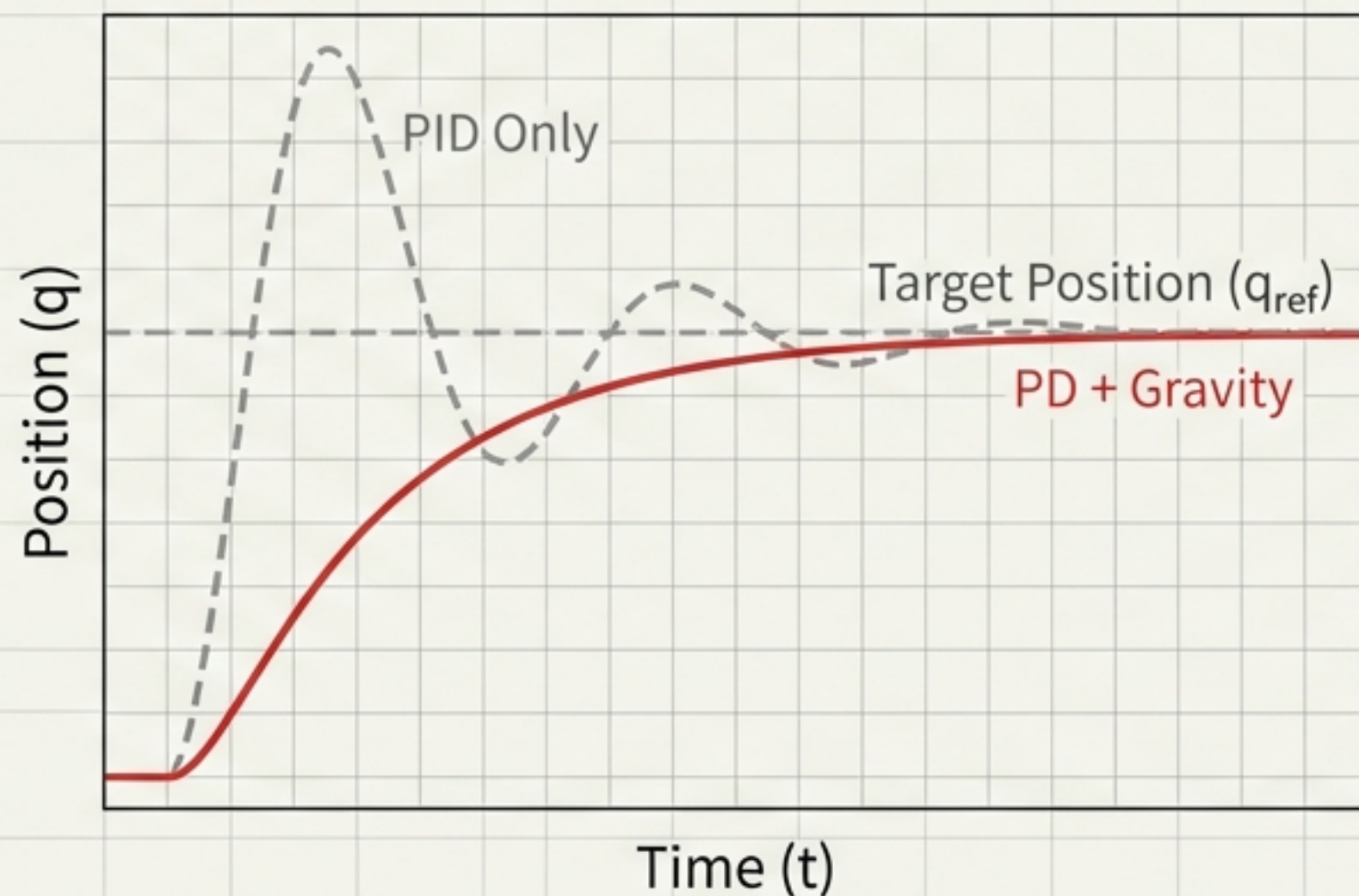
정상 상태 오차 0으로 수렴
(Converges to Zero Error)

PD 제어 + 중력 보상

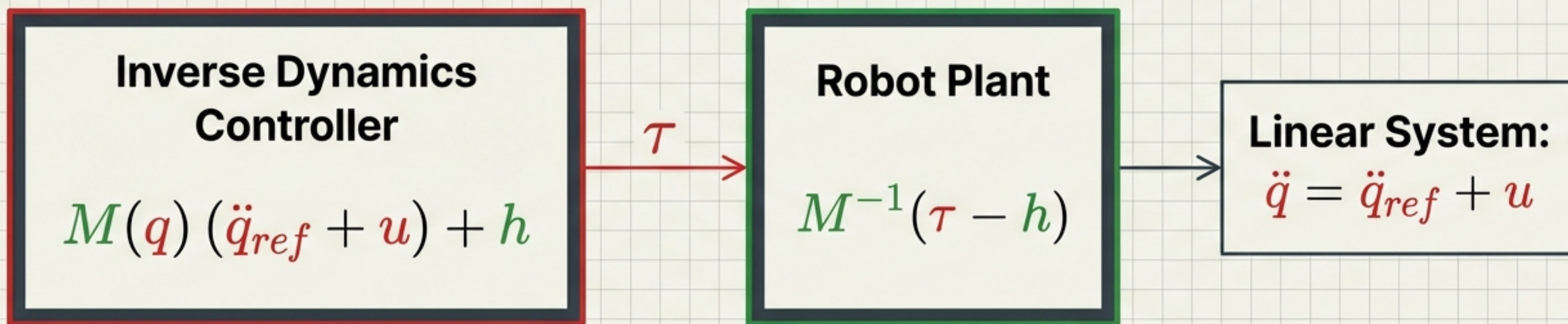
$$\tau = K_P \mathbf{e} + K_D \dot{\mathbf{e}} + \mathbf{g}(\mathbf{q})$$

중력(Gravity)은 로봇 동역학의
가장 지배적인 요소

전역 점근적 안정성
(Global Asymptotic Stability)



역동역학 제어



$$\tau = M(q)(\ddot{q}_{ref} + u) + h(q, \dot{q})$$

선형 오차 동역학 유도

1. 대입 (Substitution): $M(q)\ddot{q} + h = M(q)(\ddot{q}_{ref} + u) + h$

2. 상쇄 (Cancellation): $\cancel{M(q)\ddot{q}} = \cancel{M(q)}(\ddot{q}_{ref} + \cancel{u})$

3. 결과 (Result): $\ddot{q} = \ddot{q}_{ref} + u$

4. 피드백 설계 (Feedback Design): $u = K_D\dot{e} + K_P e$

5.

$$\ddot{e} + K_D\dot{e} + K_P e = 0$$

완벽한 2차 선형 시스템 (Perfect Linear 2nd Order System)

역동역학 제어의 변형

$$\tau = M(\ddot{q}_{ref} + K_D \dot{e} + K_P e) + h \quad (\text{Standard})$$

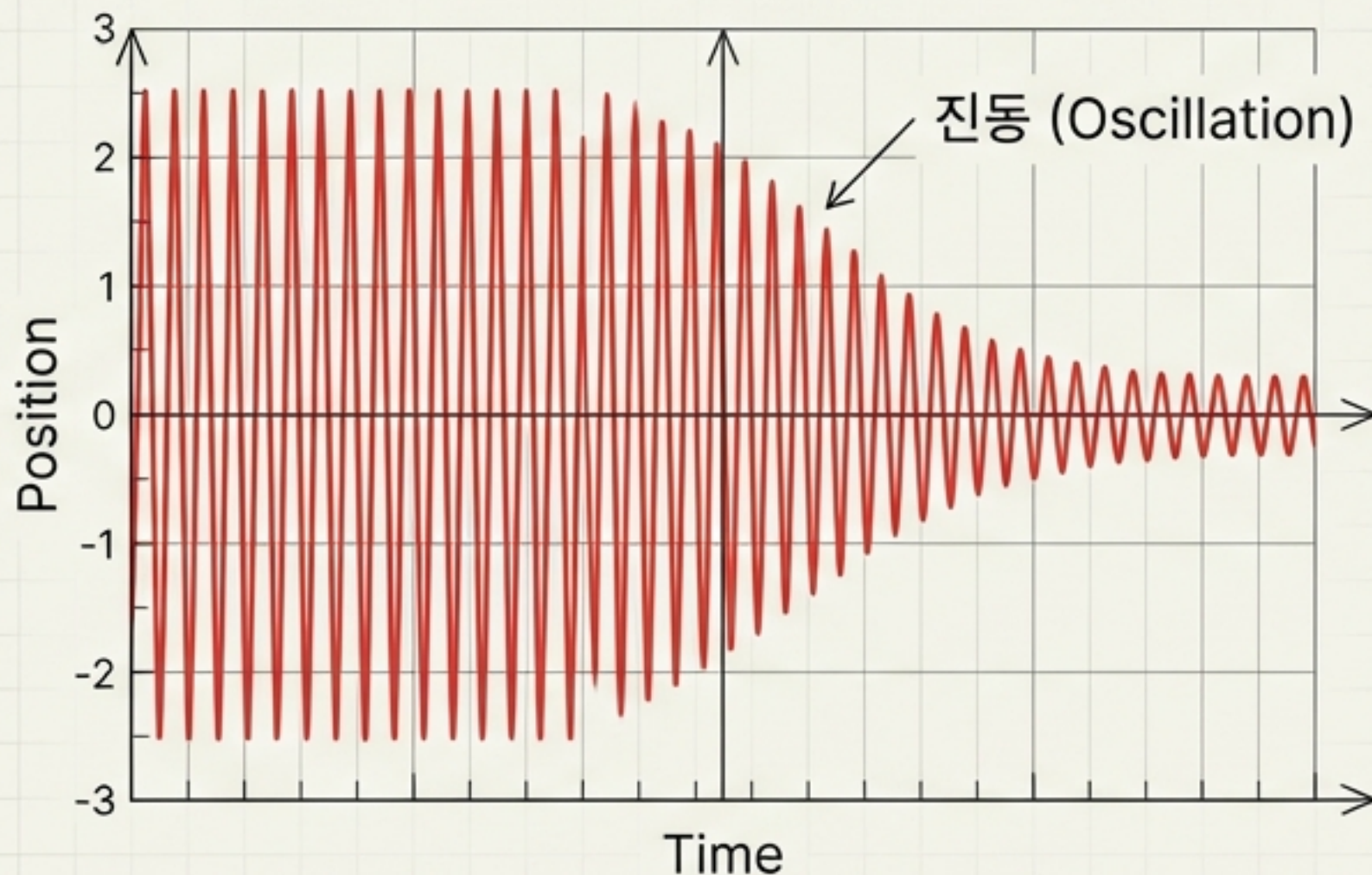
$$\tau = M\ddot{q}_{ref} + \underbrace{\bar{K}_D \dot{e} + \bar{K}_P e}_{\text{PD terms outside Mass Matrix}} + h \quad (\text{Robust Variant})$$

PD 항이 질량 행렬 외부에 위치
(PD terms outside Mass Matrix)

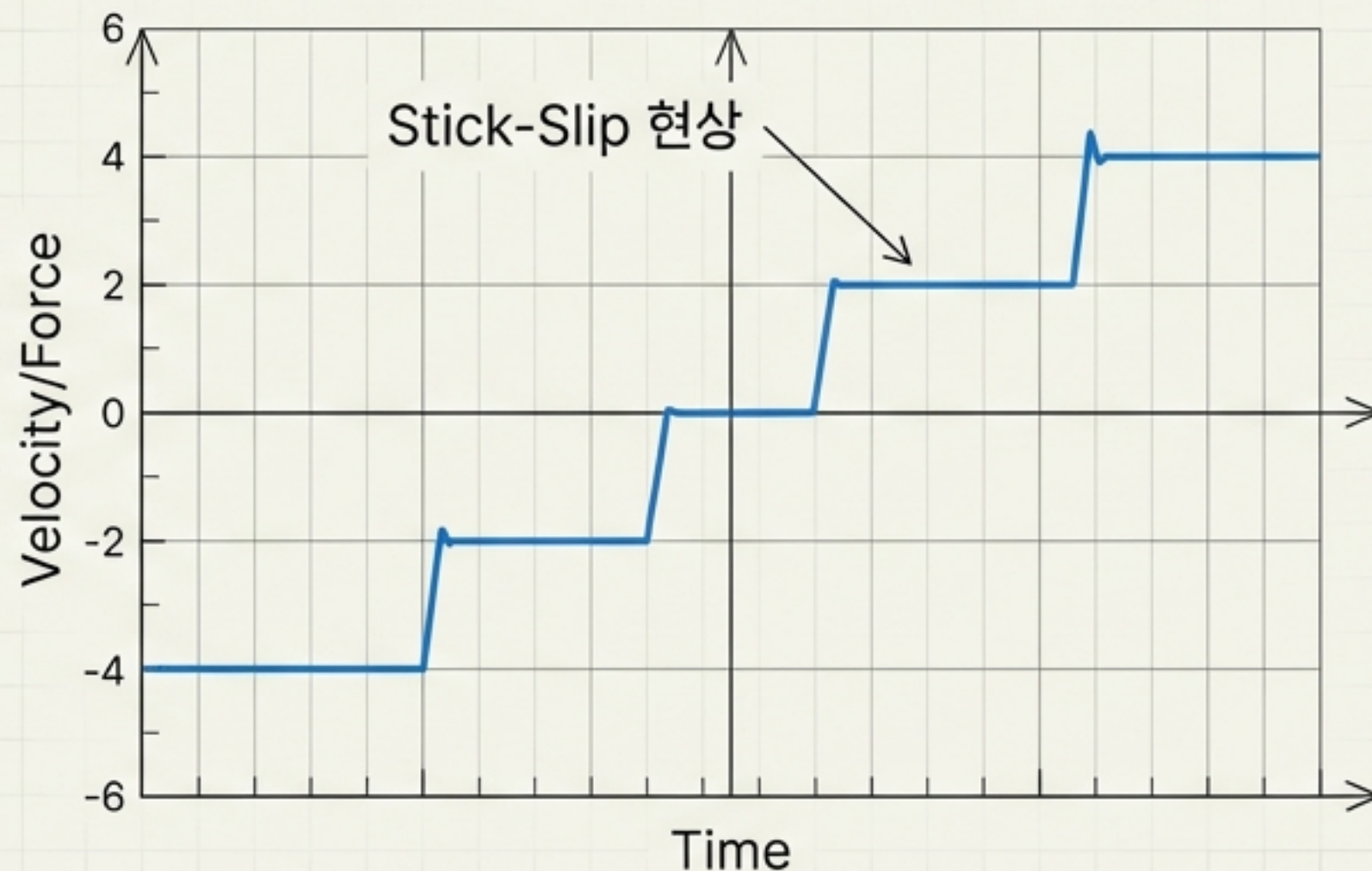
- 강인성 (Robustness): 모델 불확실성에 더 강함.
- 안정성 (Stability): $\bar{K}_P, \bar{K}_D$ 가 양의 정부호(Positive Definite)면 보장됨.

시뮬레이션 및 실제 적용 (Simulation Insights)

High Gains + Slow Loop



Friction Effects



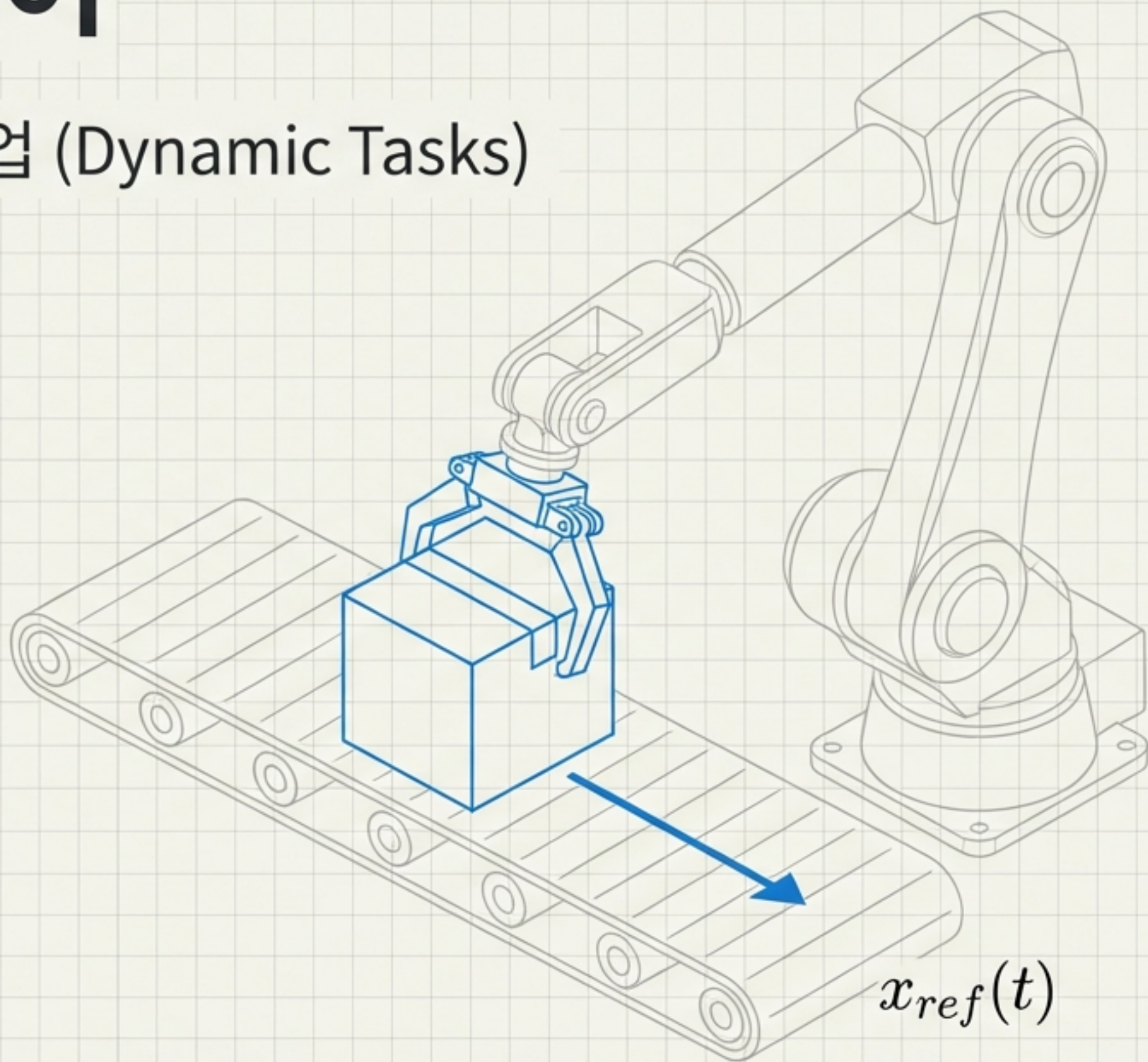
- Key Takeaways:

- 높은 게인 + 느린 제어 주기 = 불안정 (Instability)
- 마찰 보상 필수 (Friction Compensation Needed)
- 제어 주파수(Frequency)가 성능의 핵심

카르테시안 공간 제어

Motivation: 동적인 환경에서의 작업 (Dynamic Tasks)

Shift: $q \rightarrow x$



작업 공간 동역학 유도 I

Step 1

$$M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h} = \boldsymbol{\tau}$$

Pre-multiply by JM^{-1}

$$JM^{-1}M\ddot{\mathbf{q}} + JM^{-1}\mathbf{h} = JM^{-1}\boldsymbol{\tau}$$

$$J\ddot{\mathbf{q}} + JM^{-1}\mathbf{h} = JM^{-1}\boldsymbol{\tau}$$

가속도 변환 준비
(Preparing for Acceleration Transform)

작업 공간 동역학 유도 II

Focus on Kinematics

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

1

Differentiation

$$\ddot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\ddot{\mathbf{q}} + \dot{\mathbf{J}}(\mathbf{q})\dot{\mathbf{q}}$$

2

Rearranging for $\mathbf{J}\ddot{\mathbf{q}}$

$$\mathbf{J}\ddot{\mathbf{q}} = \ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}}$$

3

Substitute into previous slide's equation

$$(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}}) + \mathbf{J}\mathbf{M}^{-1}\mathbf{h} = \mathbf{J}\mathbf{M}^{-1}\boldsymbol{\tau}$$

작업 공간 동역학 유도 III

작업 공간 질량 행렬
(Operational Space Mass Matrix)

$$\Lambda(\mathbf{q}) = (\mathbf{J}\mathbf{M}^{-1}\mathbf{J}^T)^{-1}$$

$$\Lambda(\mathbf{q})\ddot{\mathbf{x}} + \mu(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{F}_c$$

$$\mu = \Lambda\mathbf{J}\mathbf{M}^{-1}\mathbf{h} - \Lambda\dot{\mathbf{J}}\dot{\mathbf{q}}$$

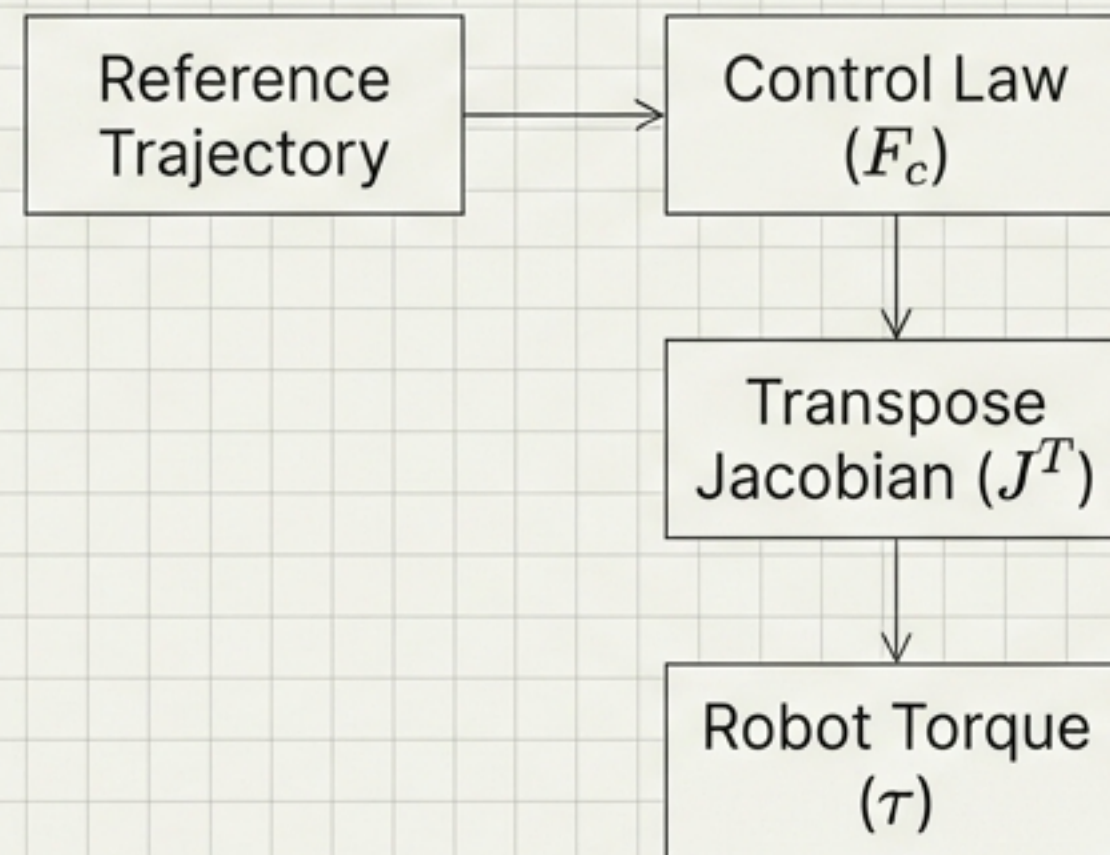
$$\boldsymbol{\tau} = \mathbf{J}^T\mathbf{F}_c$$

작업 공간 제어 법칙

$$F_c = \Lambda(q)\ddot{x}_{ref}^* + \mu(q, \dot{q})$$

$$\ddot{x}_{ref}^* = \ddot{x}_{ref} + K_D \dot{e}_x + K_P e_x$$

$$\tau = J^T F_c$$



작업 공간 선형 오차 동역학 달성 (Achieves Linear Error Dynamics in Task Space)

단순화된 카르테시안 제어

$$\tau = J^T \Lambda (\ddot{x}_{ref} + K_D \dot{e}_x + K_P e_x) + h(q, \dot{q})$$

~~$$J\dot{q} \approx 0$$~~

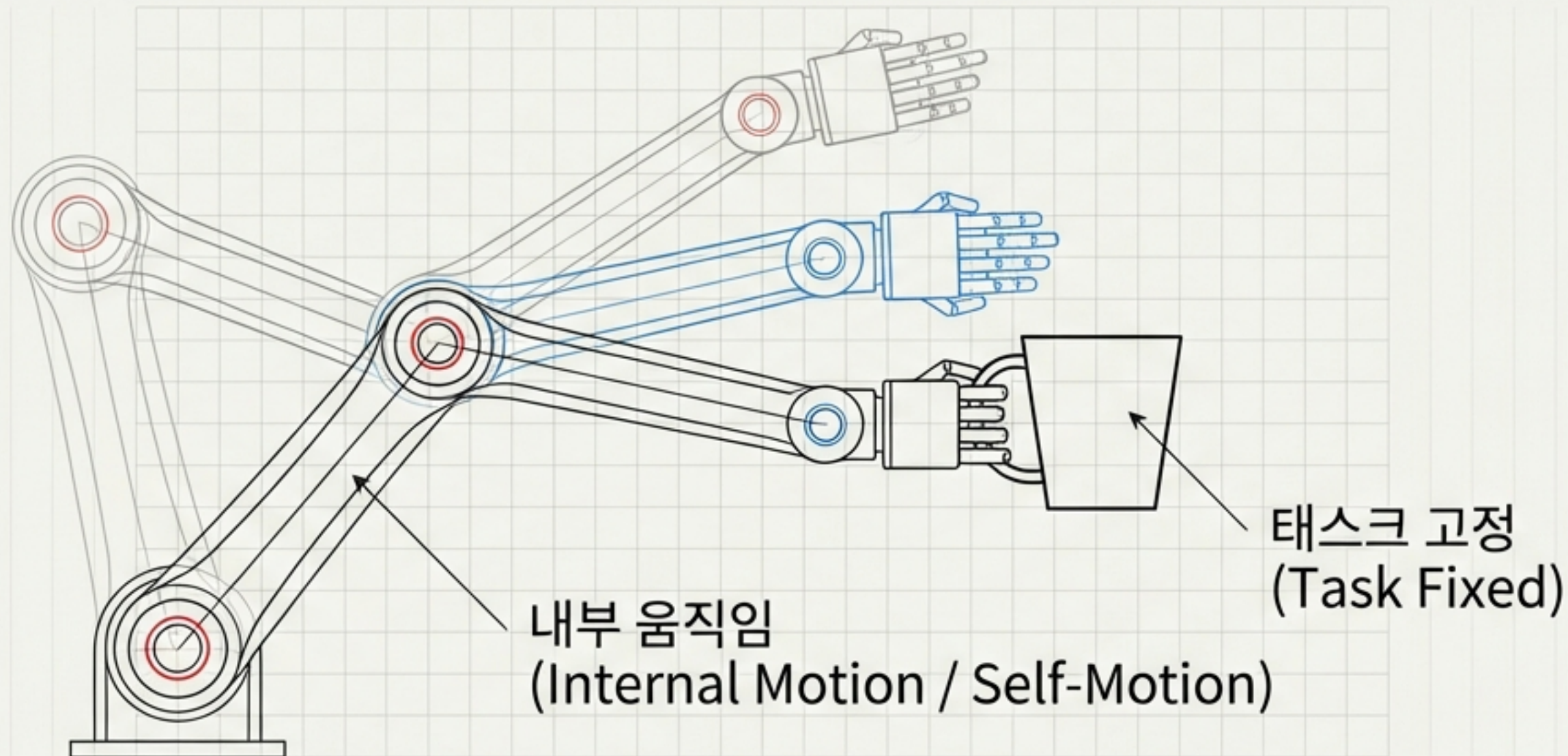
近似 (Approximation)

조인트 공간 보상으로 대체
(Replaces complex μ)

계산 효율성 (Computational Efficiency)

구현 용이성 (Ease of Implementation)

여유 자유도와 영공간 (Redundancy & Null Space)



$$n > m \text{ (Joints > Task DOFs)}$$

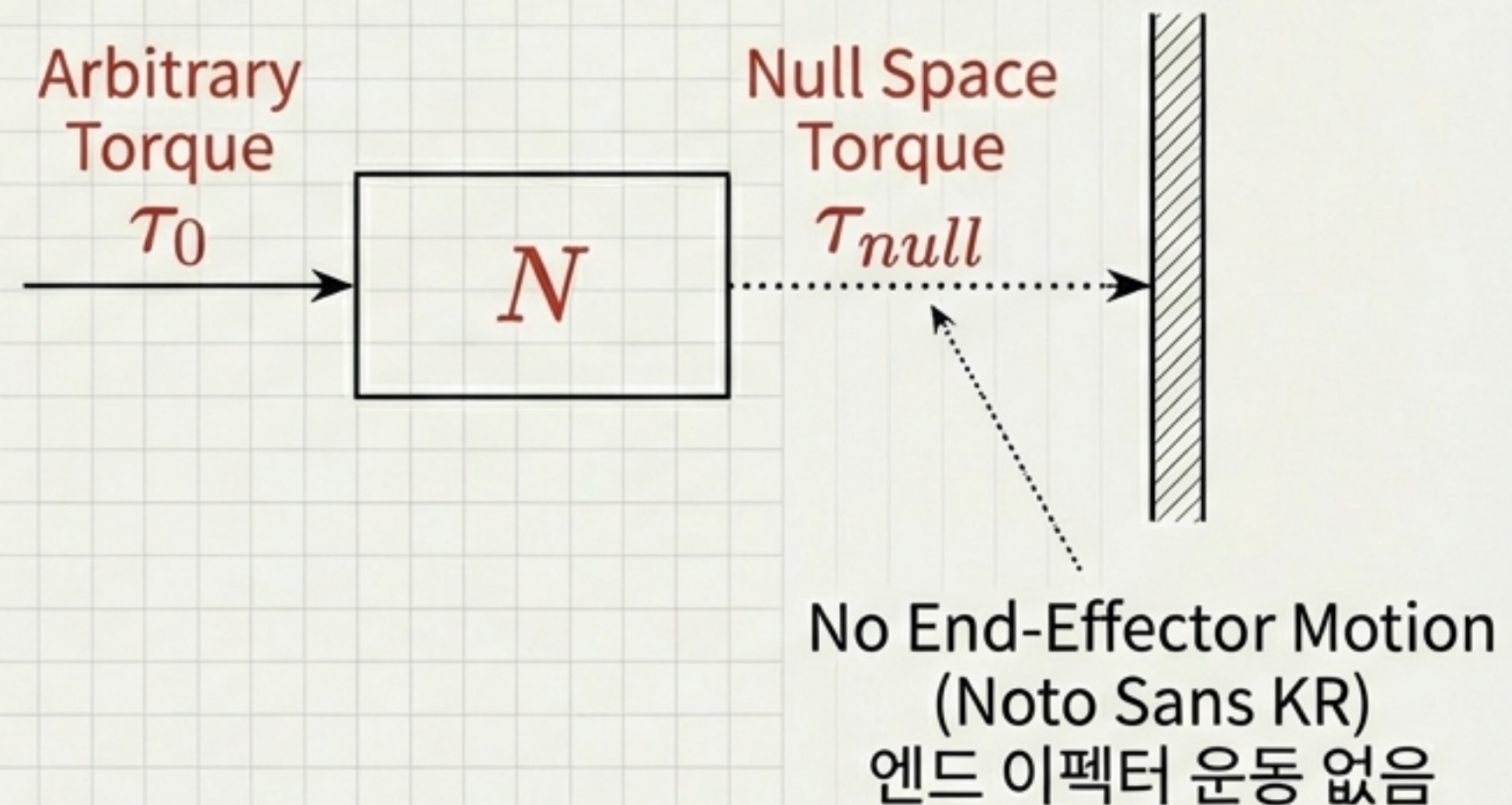
영공간 투영 행렬 (Null Space Projector)

Null Torque, defin

$$JM^{-1}\tau_{null} = 0$$

$$N = I - J^T \left(JM^{-1} J^T \right)^{-1} JM^{-1}$$

$$= I - J^T \bar{J}^T$$



영공간 성질 증명 (Proof of Null Space Property)

Check: $\mathbf{JM}^{-1}(\mathbf{N}\tau_0) = ?$

$$= \mathbf{JM}^{-1}(\mathbf{I} - \mathbf{J}^T(\mathbf{JM}^{-1}\mathbf{J}^T)^{-1}\mathbf{JM}^{-1})\tau_0$$

$$= (\mathbf{JM}^{-1} - (\mathbf{JM}^{-1}\mathbf{J}^T)(\mathbf{JM}^{-1}\mathbf{J}^T)^{-1}\mathbf{JM}^{-1})\tau_0$$

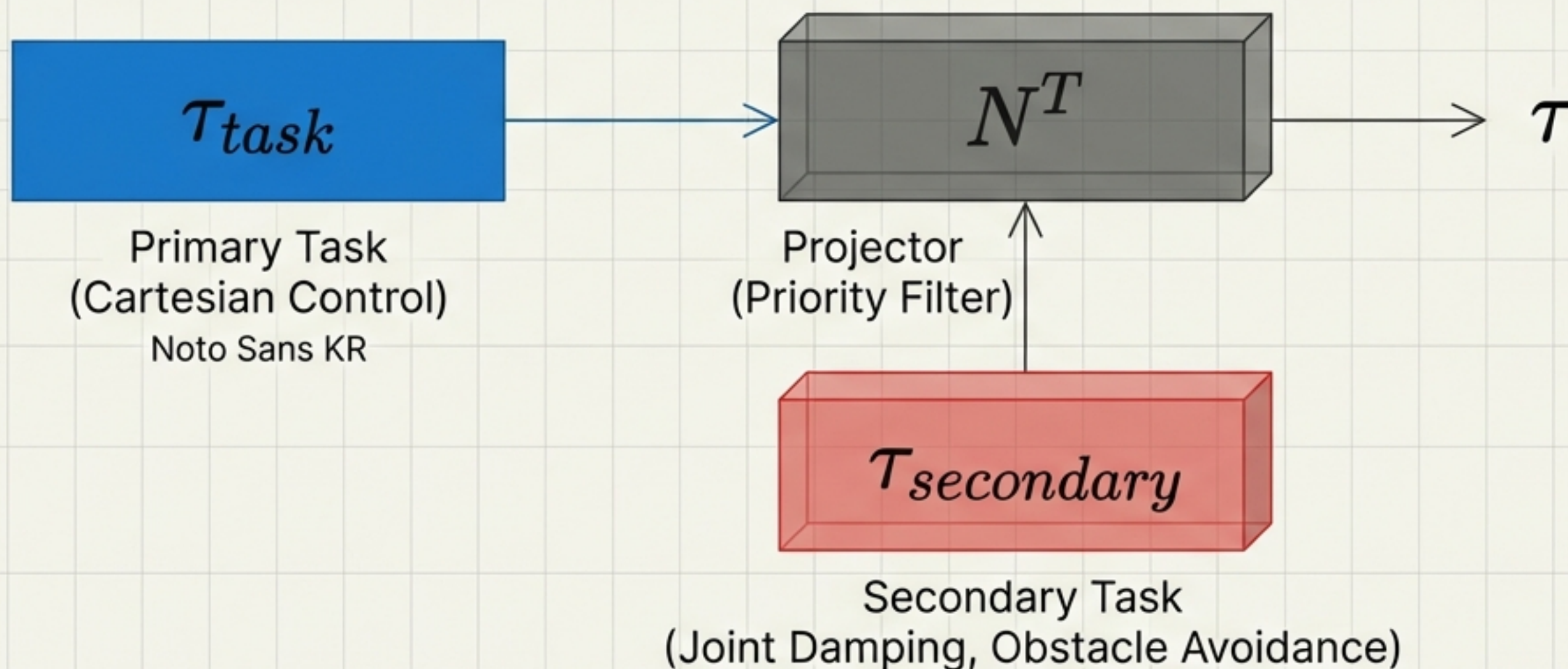
$$= (\mathbf{JM}^{-1} - \mathbf{I} \cdot \mathbf{JM}^{-1})\tau_0$$

0

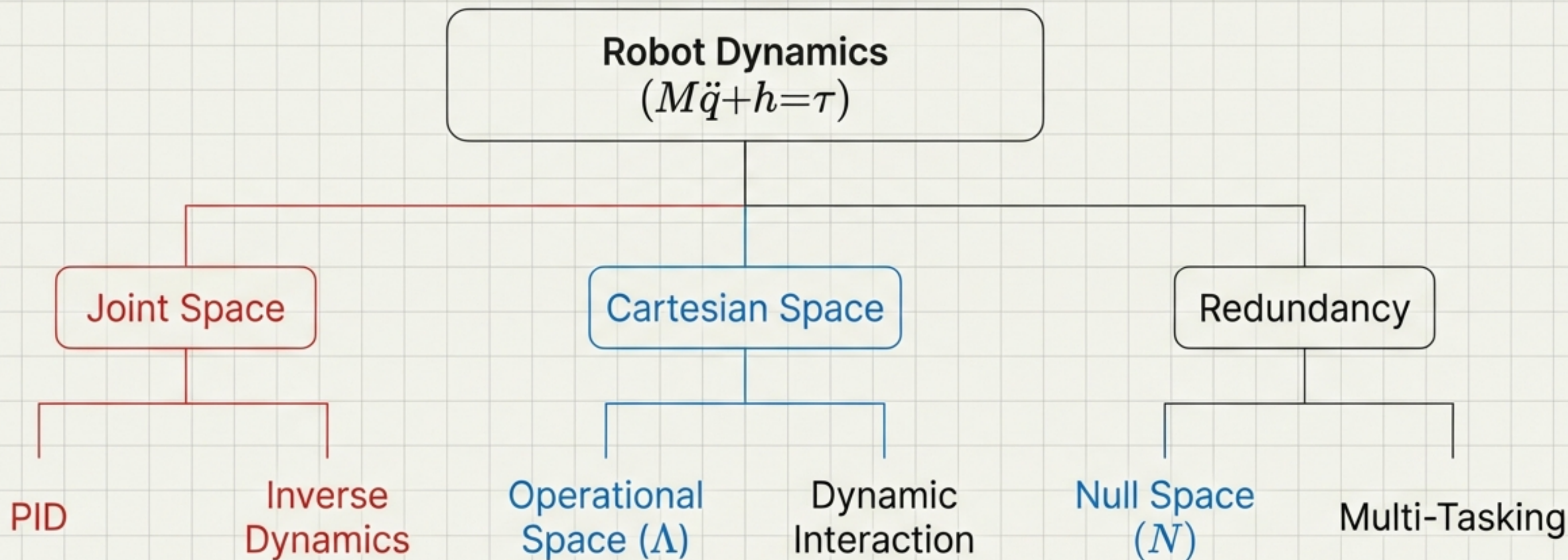
엔드 이펙터에 가속도를 유발하지 않음
(Creates Zero End-Effector Acceleration)

통합 제어 구조 (Unified Control Architecture)

$$\tau = \tau_{task} + N^T \tau_{secondary}$$



요약 및 결론 (Summary)



Next Step: 최적화 기반 제어 (Optimization-based Control)